# Religious Text Anchors and Moral Attention Reallocation in Language Models

heart-focused and cross-tradition framing as probes of motive sensitivity

Hanzhen Zhu
jennyzhu@g.harvard.edu

April 2026

**Abstract**

This paper studies a mechanistic question about moral cognition in language models: not whether a religious framing makes a model more moral overall, but whether framing changes what the model treats as morally diagnostic. The broader project uses both a generic heart-focused scaffold and a cross-tradition set of frozen study-paraphrased text anchors keyed to cited Biblical (Jewish/Christian), Buddhist, Hindu, and Islamic passages. The repository therefore has two linked artifact layers: a project-level six-condition cross-tradition readout and a narrower frozen public release artifact used for the strongest release-grade claim. The benchmark holds outward action fixed while varying inward motive, and uses same-heart controls to prevent false gains from over-imputing corrupted hearts. On the frozen 63-item public slice (23 same-act / different-motive pairs plus 40 same-heart controls), Baseline vs. Heart-focused leaves Task A flat at 0.5079, improves Task B from 0.8889 to 0.9524, raises heart-sensitivity from 0.6957 to 0.8696, and keeps same-heart control accuracy at 1.0 with heart overreach at 0.0. The corresponding exact sign test on the 23 same-act pairs remains directional rather than decisive ($p = 0.0625$ one-sided, $p = 0.125$ two-sided), while a later paired-order follow-up shows 0.0 item-level Task B order flips. On the broader project-level 63-item Qwen-1.5B-Instruct confirmation slice, the Heart-focused scaffold and Proverbs 4:23 tie for the strongest motive-sensitive improvement, Bhagavad Gita 15.15 and Qur'an 26:88–89 are smaller positive shifts, and Dhammapada 34 is null on this slice; all six conditions preserve same-heart controls, keep heart overreach at 0.0, and show clean paired-order stability on the same 23 same-act pairs. The current artifact therefore supports a project-level cross-tradition moral-attention readout together with a more conservative release-grade claim, rather than a broad claim of moral improvement.

## 1 Introduction

Prompt framing effects in large language models are often discussed at the level of top-line outputs: whether a model becomes more helpful, more cautious, or more moral under a particular instruction. For moral evaluation, that framing can be too coarse. A stronger mechanistic question is whether framing changes *what the model attends to morally*. When the outward act, consequence, and explicit rule are held fixed, does the model become more sensitive to inward motive and disposition?

This repository studies that narrower question through a benchmark design centered on *moral attention reallocation*. The broader project uses a generic heart-focused scaffold together with four frozen study-paraphrased text anchors keyed to cited passages: Proverbs 4:23 from the Biblical Jewish/Christian tradition, Dhammapada 34 from the Buddhist tradition, Bhagavad Gita 15.15 from the Hindu tradition, and Qur'an 26:88–89 from the Islamic tradition. The repository intentionally exposes two levels of evidence. At the project level, it shows a six-condition cross-tradition readout

on the same confirmation slice. At the release level, it keeps a narrower frozen artifact whose strongest supported claim is still Baseline vs. Heart-focused. The claim is intentionally modest: the current evidence does *not* show that any one religiously inflected framing makes language models more moral overall. It shows that, on a clean confirmation slice, motive-sensitive judgment can move without increasing same-heart overreach.

Two benchmark design choices are central. First, the main target items are same-act / different-motive pairs, adapted from Moral Stories [1]. These items hold outward behavior fixed while varying written motive. Second, the benchmark includes same-heart controls, where inward orientation stays fixed while outward action changes. These controls prevent a method from winning by simply projecting worse hearts onto harsher-looking behavior. This emphasis on explanation structure and moral salience also connects to prior work showing that language models can adapt moral rationalizations to framing cues [2].

Intuitively, the benchmark is asking whether the model continues to read a case mostly from its visible surface, or whether it becomes more likely to notice the moral importance of the heart from which the act flows. If two people perform the same outwardly helpful act, many human readers still distinguish between sincere care and self-display, vanity, or hypocrisy. The question here is whether framing makes that inward distinction more salient to the model without rewarding indiscriminate suspicion.

# 2 Benchmark Logic

## 2.1 Two complementary item types

**Same-act / different-motive pairs.** These are the key mechanistic items. Each pair keeps the action fixed but varies the inward orientation. A stylized example is:

- Case A: a student helps a classmate mainly to look generous.
- Case B: the same student performs the same helpful act out of sincere concern.

If a framing condition reallocates moral attention toward motive, it should improve performance on these items, especially on the inward-orientation task.

The intuition behind these items is that they separate *what was done* from *the moral orientation from which it was done*. A model that mainly tracks behavioral surface may treat the two cases as nearly interchangeable. A model that becomes more motive-sensitive should be more willing to say that the vain or resentful version reveals the worse inward orientation even when the visible action is the same.

**Same-heart controls.** These are the guardrail items. They keep inward orientation fixed while outward surface changes. A stylized example is:

- Case A: a person offers a gentle correction in order to help.
- Case B: a person offers a blunt correction with the same helping motive.

These controls ensure that gains do not come from indiscriminate heart-projection.

These controls matter because otherwise an apparent gain could be fake. A method could look better on motive-sensitive items simply by learning a crude shortcut such as "blunter act means worse heart." Same-heart controls check that the model can still preserve matched inward orientation when outward style differs.

### 2.2 Three task layers

Each pair is evaluated through three task layers:

- **Task A: overall verdict.** Which case is more morally problematic overall?
- **Task B: inward orientation.** Which case reveals a worse inward orientation?
- **Task C: reason focus.** Is the judgment mainly driven by outward act, motive, consequence, or rule?

Task B is the central mechanistic target. Task A shows whether top-line verdicts move at all. Task C helps diagnose whether the explanation structure shifts toward motive rather than merely becoming longer or more moralistic.

### 2.3 Inference-only prompt pipeline

No model is trained or fine-tuned in this repository. All reported numbers come from prompting-only local inference through the repository's multi-pass runner (`run_transformers_multipass.py`) and the shared `pilot_v12` prompt package.

The released confirmation artifacts use the same multi-pass structure:

- **Task A / Task C pass:** the model sees the two full case texts and returns an overall verdict, a single reason label, and a short explanation.
- **Task B intention source:** the released configs use benchmark-summary copy mode, so the intention-only comparison is built from the benchmark's written `motive_summary` fields rather than a model-generated copy step.
- **Relation gate:** the model first judges whether the two written intentions express the same or different inward orientation.
- **Task B pass:** only if the relation gate returns *different* does the model choose which intention reflects the worse inward orientation; otherwise Task B is set to *Same*.
- **Order handling:** the released configs use canonical-source order remapping, so first/second intention responses are mapped back to canonical case sources before scoring.

The six-condition prompt family keeps this pipeline fixed. Baseline uses no added frame. Heart-focused adds a generic inward-motive scaffold. Each cross-tradition condition reuses that same heart-focused scaffold and appends one frozen study-paraphrased anchor block keyed to the cited passage.

### 2.4 Metrics and robustness checks

The public artifact reports seven first-class metrics. Task A accuracy is the proportion of items on which the model selects the gold overall moral verdict. Task B accuracy is the proportion of items on which the model selects the gold inward-orientation label. Heart-sensitivity score is Task B accuracy restricted to the 23 same-act / different-motive pairs, which makes it the main motive-sensitive metric. The reason-focus metric "$P(\text{reason} = \text{motive})$" is the proportion of items on which Task C selects *motive* as the primary reason. Same-heart control accuracy and heart overreach rate are both measured on the 40 same-heart controls; the former tracks correct *Same* judgments and the

latter tracks false projections of worse inward orientation. Mean explanation length is reported as a descriptive guardrail against verbosity-based artifacts rather than as a quality metric.

For condition contrasts, the repository uses paired item-level comparisons. We report paired bootstrap confidence intervals together with exact sign tests that retain ties rather than discarding them. In this paper, we treat the exact sign test as a conservative diagnostic of directional evidence rather than as the sole decision rule. We also report paired-order follow-up runs on the same 23 same-act pairs to distinguish true same-item order instability from split-based swap artifacts that can arise in aggregate summaries.

## 3 Frozen Public Release Artifact

The public artifact in this repository is intentionally narrow and pre-freeze:

- **Model:** Qwen-1.5B-Instruct
- **Conditions:** Baseline vs. Heart-focused
- **Slice size:** 63 items
- **Composition:** 23 same-act motive pairs + 40 same-heart controls
- **Prompt package:** `prompts/pilot_v12` with benchmark-summary Task B copy mode and canonical-source Task B order remapping
- **Decoding:** temperature = 0, top-$p$ = 1.0, $n$ = 1, max_new_tokens = 120, max_attempts = 2

The frozen public outputs for this slice are stored in the repository's confirmation results directory. The full 160-item main benchmark is not yet frozen. Later project-level readouts rerun this same 63-item slice with a broader prompt family, but those reruns are not the canonical release-grade claim.

## 4 Results

Figure 1 shows the narrower frozen release result. Task A stays flat. The strongest movement is in Task B and heart-sensitivity score. Reason focus also shifts modestly toward motive. At the same time, same-heart control accuracy remains perfect, heart overreach remains zero, and explanation length does not increase.

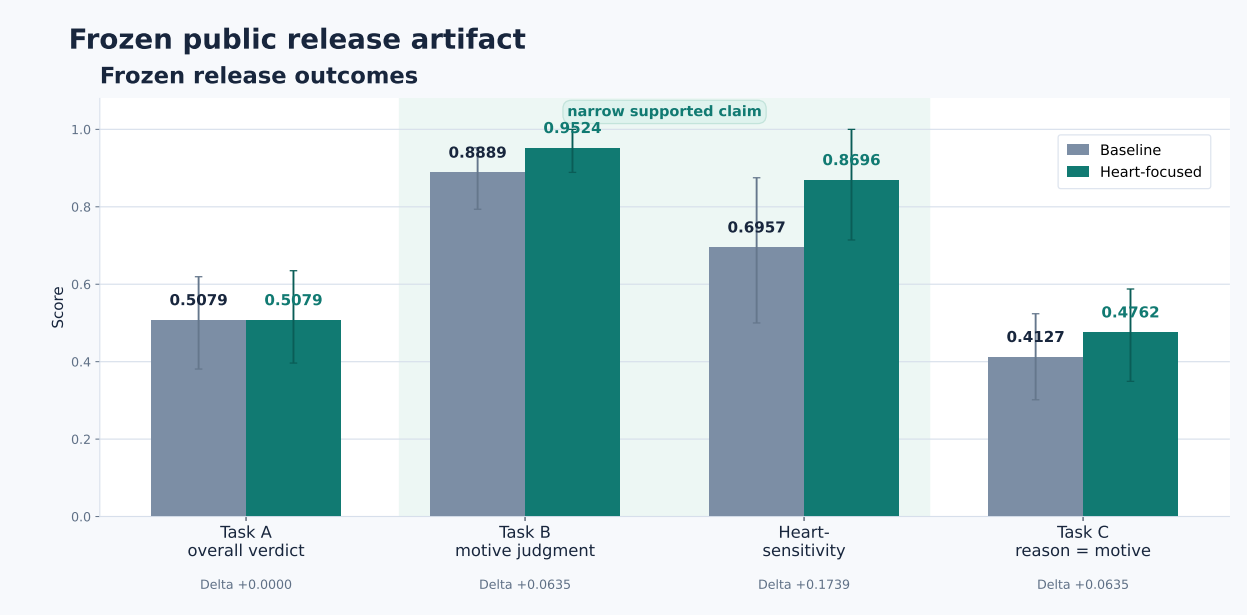


**Figure 1:** Frozen public release artifact, comparing Baseline and Heart-focused framing. The chart shows the primary release-grade outcomes on the confirmation slice. Task A remains flat, while Task B and heart-sensitivity move upward, and Task C shifts modestly toward motive.

**Table 1:** Main public confirmation metrics.

| Metric | Baseline | Heart-focused | Delta |
|---|---|---|---|
| Task A accuracy | 0.5079 | 0.5079 | +0.0000 |
| Task B accuracy | 0.8889 | 0.9524 | +0.0635 |
| Heart-sensitivity score | 0.6957 | 0.8696 | +0.1739 |
| P(reason = motive) | 0.4127 | 0.4762 | +0.0635 |
| Same-heart control accuracy | 1.0000 | 1.0000 | +0.0000 |
| Heart overreach rate | 0.0000 | 0.0000 | +0.0000 |
| Mean explanation characters | 111.54 | 109.16 | -2.38 |

The release pattern is asymmetric in a way that matters for interpretation. Task B improves by +0.0635 with a paired bootstrap interval of [0.0159, 0.1270], and heart-sensitivity improves by +0.1739 with a paired bootstrap interval of [0.0475, 0.3448]. By contrast, Task A remains exactly flat, the Task C motive-focus shift is smaller and less stable (+0.0635 with interval [-0.0159, 0.1429]), and mean explanation length decreases slightly rather than inflating. That pattern is more consistent with motive-sensitive attention reallocation than with a general increase in moralistic verbosity.

One intuitive way to read this pattern is that the model is not obviously becoming harsher or more morally serious in general. Instead, it is becoming somewhat more likely to treat motive as the thing that breaks a tie between two outwardly similar acts. In other words, the effect shows up more clearly in *what feature the model treats as diagnostic* than in a broad shift in overall condemnation.

The exact paired test on the 23 same-act motive pairs remains conservative:

- Better / worse / tie counts for heart-sensitivity: 4 / 0 / 19
- Exact sign test: $p = 0.0625$ one-sided, $p = 0.125$ two-sided

So the current public result is best described as *directional confirmation*. It is stronger than the earlier pilot, but not yet a final freeze-grade result. A later paired-order follow-up on the same 23 same-act items found 0.0 item-level Task B order flips and 0.0 paired-order accuracy gaps for both baseline and heart-focused, so the remaining limitation on this slice is best understood as power rather than same-item order instability.

## 5 Interpretation

The public artifact supports the following claim:

> heart-focused framing may reallocate moral attention toward inward motive on a clean confirmation slice, without increasing same-heart overreach.

This wording matters for two reasons.

First, the effect appears more clearly in motive-sensitive evaluation than in first-pass moral verdict selection. Task A remains unchanged, while Task B and heart-sensitivity improve. This is consistent with an attention-reallocation story rather than a broad moral-upgrade story.

Second, the guardrail holds. Same-heart control accuracy remains 1.0 and heart overreach remains 0.0 in both conditions. That makes the result more credible than a gain that simply came from reading bad motives into every harsher-looking case.

Stated more plainly, the current effect looks like this: the model does *not* become uniformly more likely to condemn a case, but it does become more likely to say that two outwardly matched acts can still differ morally because they flow from different inner motives. That is the specific sense in which we use the phrase *moral attention reallocation*.

## 6 Limitations

The current public artifact has several explicit boundaries.

- It is a pre-freeze confirmation slice, not the full main benchmark.
- The canonical public comparison is Baseline vs. Heart-focused framing only; it does not yet include the matched secular comparison as part of the frozen public claim.
- The result is directional rather than decisive under conservative paired testing.
- The broader transformed Moral Stories benchmark is not yet fully double-annotated and frozen.
- The six-condition cross-tradition readout is currently single-model and should be read as exploratory robustness evidence rather than as a finalized cross-model benchmark result.
- The four anchored conditions are controlled text-anchor probes layered on one shared scaffold, not comprehensive evaluations of religious traditions, translations, or theological systems.

These limitations are part of the intended honesty of the artifact. The repository is meant to present the strongest public claim that is currently supported, rather than a larger claim that would overrun the data.

## 7 Project-Level Cross-Tradition Readout

Beyond the frozen release artifact, the repository contains a broader project-level cross-tradition readout that keeps the same benchmark logic but expands the prompt family to six conditions: *baseline*, *heart-focused*, and four cross-tradition religious text anchors: *Proverbs 4:23* from the Biblical Jewish/Christian tradition, *Dhammapada 34* from the Buddhist tradition, *Bhagavad Gita 15.15* from the Hindu tradition, and *Qur'an 26:88–89* from the Islamic tradition. These four anchors are implemented as frozen study paraphrases inserted into the same heart-focused scaffold rather than as full tradition-specific prompt rewrites. This readout is not part of the frozen release-grade claim. Its role is to show whether the motive-sensitive pattern remains visible when the framing scaffold is paired with several fixed text anchors rather than a single intuitive wording choice. Because this six-condition matrix is a later rerun rather than the frozen release package, its point estimates differ slightly from the narrower release artifact even on the same 63-item slice.

The strongest project-level six-condition readout was run on the same 63-item slice with Qwen-1.5B-Instruct. The resulting pattern is heterogeneous but informative. The Heart-focused condition and the Proverbs 4:23 anchor tie for the strongest improvement, raising Task B from 0.8889 to 0.9683 and heart-sensitivity from 0.6957 to 0.9130; for both conditions, the exact sign test on the same-act pairs is $p = 0.03125$ one-sided and $p = 0.0625$ two-sided. Bhagavad Gita 15.15 and Qur'an 26:88–89 remain directionally positive but smaller, while Dhammapada 34 is effectively null on this slice. All six conditions preserve same-heart control accuracy at 1.0 and keep heart overreach at 0.0.

The paired-order follow-up matters as much as the point estimates. A confirmation paired-order pack on the 23 same-act items shows 0.0 item-level Task B order flips and 0.0 paired-order Task B gaps across all six conditions. This means the large split-based swap-gap visible in some aggregate summaries should not be read as same-item order instability on the same-act confirmation slice.

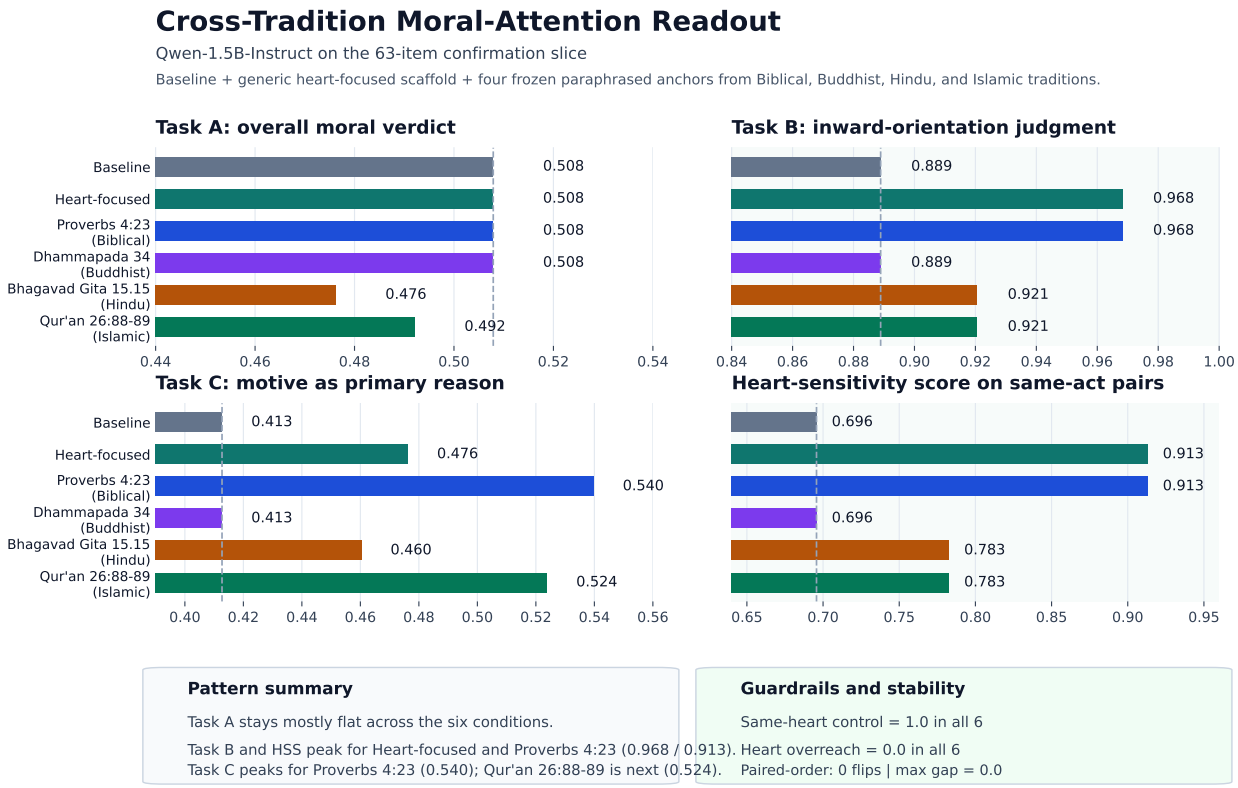


**Figure 2:** Project-level six-condition confirmation on Qwen-1.5B-Instruct, including generic heart-focused framing plus four frozen study-paraphrased anchors keyed to Biblical (Jewish/Christian), Buddhist, Hindu, and Islamic passages. The four panels report Task A overall moral verdict, Task B inward-orientation judgment, Task C motive-as-primary-reason, and heart-sensitivity on same-act pairs. The strongest gains on the confirmation slice come from the Heart-focused prompt and the Proverbs 4:23 anchor from the Biblical Jewish/Christian tradition, which tie on Task B and heart-sensitivity score. All six conditions preserve same-heart controls, heart overreach stays at zero, and the confirmation paired-order diagnostic shows no item-level Task B order flips.

Table 2 gives the main reader-facing comparison in one place: one metric-by-condition matrix for horizontal comparison across all six conditions. This keeps the project-level readout compact and makes the cross-condition pattern easy to scan without relying on the figure alone.

**Table 2:** Project-level six-condition metric-by-condition matrix on Qwen-1.5B-Instruct. Task A, Task B, and Task C use all 63 confirmation items; HSS and paired-order Task B use the 23 same-act pairs.

| Metric | Baseline<br>No religious text | Heart-focused<br>Generic scaffold | Proverbs 4:23<br>Biblical (Jewish/Christian) | Dhammapada 34<br>Buddhist | Bhagavad Gita 15.15<br>Hindu | Qur'an 26:88-89<br>Islamic |
|---|---|---|---|---|---|---|
| Task A accuracy (overall verdict) | 0.5079 | 0.5079 | 0.5079 | 0.5079 | 0.4762 | 0.4921 |
| Task B accuracy (inward orientation) | 0.8889 | 0.9683 | 0.9683 | 0.8889 | 0.9206 | 0.9206 |
| Task C motive-focus rate | 0.4127 | 0.4762 | 0.5397 | 0.4127 | 0.4603 | 0.5238 |
| Heart-sensitivity score (same-act) | 0.6957 | 0.9130 | 0.9130 | 0.6957 | 0.7826 | 0.7826 |
| Same-heart control accuracy | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 |
| Heart overreach rate | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| Mean explanation length (chars) | 112.9 | 105.7 | 108.0 | 106.6 | 114.7 | 109.0 |
| Paired-order Task B accuracy | 0.6957 | 0.9130 | 0.9130 | 0.6957 | 0.7826 | 0.7826 |
| Order-flip rate | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| Paired-order Task B gap | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |

This project-level readout does not justify a new freeze-grade cross-model claim. The earlier pilot remained heterogeneous across models, and the six-condition pattern should still be interpreted

more conservatively than the narrower release artifact. It is also important not to overread the four anchor conditions as a head-to-head comparison of whole traditions: because all four anchors are layered on top of the same heart-focused scaffold, the design isolates the effect of a fixed additional text block rather than providing a fair comparison of traditions, translations, or doctrines. Within that narrower interpretation, however, the readout does strengthen the overall research story in one important way: the core motive-sensitive effect on Qwen-1.5B-Instruct is no longer dependent on only one prompt wording. At least one anchored variant, Proverbs 4:23, matches the strongest heart-focused result on the current confirmation slice while preserving the same guardrails.

## 8 Artifact Availability

The repository includes:

- the public confirmation outputs,
- the figure-generation scripts,
- the reproduction script for the Qwen-1.5B confirmation slice,
- and this paper in both PDF and source form.

Repository: https://github.com/hanzhenzhujene/moral-attention-reallocation

## References


[1] Denis Emelin, Ronan Le Bras, Jena D. Hwang, Maxwell Forbes, and Yejin Choi. 2021. Moral Stories: Situated Reasoning about Norms, Intents, Actions, and their Consequences. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, pages 698–718. Association for Computational Linguistics. URL: https://aclanthology.org/2021.emnlp-main.54/

[2] Gabriel Simmons. 2023. Moral Mimicry: Large Language Models Produce Moral Rationalizations Tailored to Political Identity. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 4: Student Research Workshop)*, pages 282–297. Association for Computational Linguistics. URL: https://aclanthology.org/2023.acl-srw.40/